

THE STEENROD ALGEBRA AND ITS DUAL FOR CONNECTIVE K-THEORY^(*)

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A basic spectrum of modern algebraic topology is the spectrum for connective real K-theory, b_0 . Its cohomological structure was described by Adams [1] and Stong [10] at the prime 2 and W. Singer [9] at odd primes. In this note we will only be concerned with 2 where

$$H^*(b_0, \mathbb{Z}/(2)) \cong \mathcal{A}(2)/\mathcal{A}(2)\{Sq^1, Sq^2\},$$

and $\mathcal{A}(2)$ is the mod (2) Steenrod algebra.

Mahowald in [6], [7] and Don Anderson (unpublished) initiated the study of the Steenrod algebra for connective K-theory $\mathcal{A}(b_0) = \{b_0, b_0\}_*$ and $\pi_*(b_0 \wedge b_0)$. This latter is useful (indeed essential) for constructing and studying the b_0 generalized Adams spectral sequence [3]. Actually though what is most needed are not the homotopy groups but the homotopy type.

In general, if $\mathcal{M}$ is a spectrum, then the techniques of "classical" homological algebra are applicable to the study of the $\mathcal{M}$ -generalized Adams spectral sequence only if

$$\mathcal{M} \wedge \mathcal{M} \simeq \bigvee_J \Sigma^J(\mathcal{M}).$$

^(*) This research is supported in part by NSF grant GP 43967.

For example, the Eilenberg-MacLane spectra $K(\mathbb{Z}/p, 0)$, and the Brown-Peterson spectra have this property. It is easily checked though that $b_0 \wedge b_0$ cannot split in this way (already true by the 7 skeleton) so comparatively little attention has been given to the resulting Adams spectral sequence.

In this note we obtain the homotopy type of $b_0 \wedge b_0$ at the prime 2. It turns out to be associated not to b_0 alone but to b_0, b_{sp} and certain spaces obtained from b_0, b_{sp} by using $K(\mathbb{Z}/2, 0)$ resolutions of b_0, b_{sp} . (Here b_{sp} is the 3 connective cover of b_0 , and again from [10])

$$H^*(b_{sp}, \mathbb{Z}/2) \cong \mathcal{A}(2) / \mathcal{A}(2) \{Sq^1, Sq^5\}.$$

In terms of these spaces whose precise definitions will be given in 2 our main theorem is

$$\begin{aligned} \text{Theorem A. } b_0 \wedge b_0 &\simeq \bigvee_J \Sigma^{J \cdot K(\mathbb{Z}/2, 0)} \vee b_0 \vee \Sigma^4 b_{sp} \vee \Sigma^8 b_0^{(3)} \\ &\vee \dots \vee \Sigma^{8k-4} b_{sp}^{2k-1-\alpha(k-1)} \vee \Sigma^{8k} b_0^{2k-\alpha(k)} \vee \dots \end{aligned}$$

This decomposition is actually explicit. Let ψ^3 be the Adams operation $\psi^3: B_0 \rightarrow B_0, \psi^3|_{B_0[8k, 8k+1, \dots]}$ and looped $8k$ times is $9^k \psi^3$; hence localizing at 2, so 3 is a unit, we have $\frac{1}{9^k} \psi^3|_{B_0[8k, \dots]}$ define an operation in the b_0 spectrum which we also denote ψ^3 . Consider the polynomial

$$\phi_n = (\psi^3 - 1)(\psi^3 - 9) \dots (\psi^3 - 9^{n-1}).$$

$\phi(k) = \#1$ in binomial coeffs

We prove

Theorem B. Let $b_0^{(n)}$ be the space which is universal for maps $X \rightarrow b_0$ which are trivial with respect to all higher $\mathbb{Z}/2$ cohomology operations of order $< n$; then

(i) ϕ_{2n} admits a factorization

$$b_0 \xrightarrow{\theta_{2n}} \Sigma^{8n} b_0^{(2n-\alpha(n))} \xrightarrow{\lambda} \Sigma^{8n} b_0 \xrightarrow{j} b_0$$

where λ is the universal map and j the usual inclusion given by Bott periodicity.

(ii) ϕ_{2n+1} admits a factorization

$$b_0 \xrightarrow{\theta_{2n+1}} \Sigma^{8n+4} b_{sp}^{(2n-\alpha(n))} \xrightarrow{\lambda} \Sigma^{8n+4} b_{sp} \xrightarrow{\hat{j}} \Sigma^{8n} b_0 \xrightarrow{j} b_0.$$

The θ_n 's defined in Theorem B allow us to define maps

$$b_0 \wedge b_0 \rightarrow \begin{cases} \Sigma^{8n} b_0^{(2n-\alpha(n))} \\ \Sigma^{8n+4} b_{sp}^{(2n-\alpha(n))} \end{cases}$$

using the ring structure of b_0 and the b_0 module structure of b_{sp} .

Theorem C. $b_0^{(n)}$ and $b_{sp}^{(n)}$ are filtered modules over b_0 (i.e., there are universal maps

$$\mu : b_0^{(i)} \wedge b_0^{(n)} \rightarrow b_0^{(n+i)}$$

$$\mu : b_0^{(i)} \wedge b_{sp}^{(n)} \rightarrow b_{sp}^{(n+i)}$$

so the diagrams

$$\begin{array}{ccc} b_0^{(i)} \wedge b_{sp}^{(n)} & \longrightarrow & b_{sp}^{(n+i)} \\ \downarrow \lambda \wedge \lambda & & \downarrow \lambda \\ b_0 \wedge b_{sp} & \longrightarrow & b_{sp} \\ \downarrow 1 \wedge \hat{j} & & \downarrow \hat{j} \\ b_0 \wedge b_0 & \longrightarrow & b_0 \end{array}$$

$$\begin{array}{ccc} b_0^{(i)} \wedge b_0^{(n)} & \longrightarrow & b_0^{(n)} \\ \downarrow \lambda \wedge \lambda & & \downarrow \lambda \\ b_0^{(i)} \wedge b_0 & \longrightarrow & b_0 \end{array}$$

commute and there are minimal models for the $b_0^{(n)}, b_{sp}^{(n)}$ so that
the maps

$$b_0 \wedge b_0 \longrightarrow \Sigma^{8n} b_0^{2n-\alpha(n)}$$

defined as the composites

$$\begin{aligned} b_0 \wedge b_0 &\xrightarrow{1 \wedge \theta_{2n}} b_0 \wedge \Sigma^{8n} b_0^{2n-\alpha(n)} \xrightarrow{\mu} \Sigma^{8n} b_0^{2n-\alpha(n)} \\ b_0 \wedge b_0 &\xrightarrow{1 \wedge \theta_{2n+1}} b_0 \wedge \Sigma^{8n+4} b_{sp}^{2n-\alpha(n)} \xrightarrow{\mu} \Sigma^{8n+4} b_{sp}^{2n-\alpha(n)} \end{aligned}$$

together with (usual) maps into the Eilenberg-MacLane spectrum
 $K(\mathbb{Z}/2, 0)$ defines the splitting in Theorem A.

Remark. These theorems represent part of the foundation for ongoing work with Mahowald which will appear in due course. For now let me observe that the splitting above implies that while the techniques of classical homological algebra are not appropriate to the b_0 -spectrum we can modify homological algebra by introducing valuated or filtered categories and filtered resolutions which implies filtered Ext groups, etc. With these modifications a beautiful theory seems to emerge which contains deep information about homotopy and is easily handled, using techniques developed in ordinary homological algebra for Abelian rings.

1. Some techniques from homological algebra.

We may write $H^*(b_0, \mathbb{Z}/2) \cong \mathcal{A}(2)/\mathcal{A}(2)\overline{\mathcal{A}}_1$ where $\mathcal{A}_1$ is the subalgebra of $\mathcal{A}(2)$ generated by Sq^1 and Sq^2 . Under such circumstances we can use change of rings techniques to reduce the study of the ordinary Adams spectral sequence for spaces $X \wedge b_0$ to $\text{Ext}_{\mathcal{A}_1}^*(H^*(X), \mathbb{Z}/2)$.

Lemma 1.1 (Change of rings). Let M have the form $N \otimes (\mathcal{A}(2)/\mathcal{A}(2)\overline{B})$ where N is an $\mathcal{A}(2)$ module and B is a sub-Hopf algebra of $\mathcal{A}(2)$. Suppose, moreover, there is an $\mathcal{A}(2)$ action on B so

$$\begin{array}{ccc}
 \mathcal{A}(2) \otimes B & \longrightarrow & B \\
 \uparrow & & \uparrow s \\
 B \otimes B & \longrightarrow & B
 \end{array}$$

commutes; then $\text{Ext}_{\mathcal{A}(2)}(N \otimes \mathcal{A}(2) / \mathcal{A}(2)B, \mathbb{Z}/2) \cong \text{Ext}_B(N, \mathbb{Z}/2)$
(well known).

$\mathcal{A}_1$ satisfies the hypothesis of 1.1 since we can construct a space X with $H^*(X, \mathbb{Z}/2) \cong \mathcal{A}_1$ as a module over $\mathcal{A}_1$.
Thus $\text{Ext}_{\mathcal{A}(2)}(H^*(b_0), \mathbb{Z}/2) \cong \text{Ext}_{\mathcal{A}_1}(\mathbb{Z}/2, \mathbb{Z}/2)$ which has the form

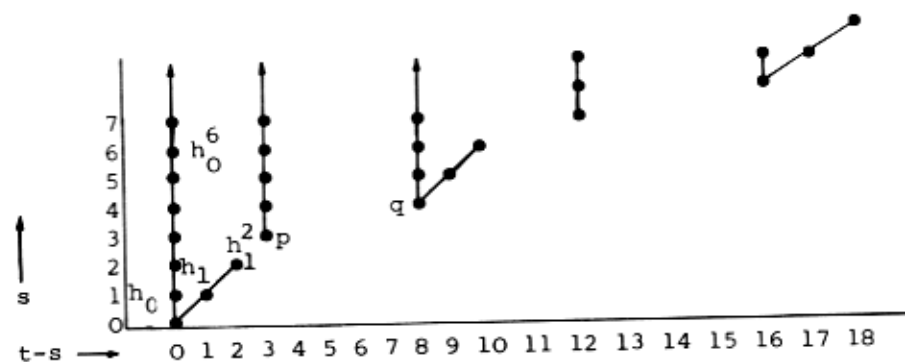


Figura 1.2

Next we recall the result of Wall [11].

Theorem 1.3. Let $Q(\mathcal{A}_1) = [Sq^1, Sq^2]$; then $Q^2 = 0$ and
if M is a connected locally finitely generated $\mathcal{A}_1$ -module,
then M is $\mathcal{A}_1$ free if and only if

$$H_*(M, Sq^1) \cong H_*(M, Q) = 0.$$

Definition 1.4. Two modules M and N over $\mathcal{A}_1$ are stably equivalent if there exist connected locally finitely generated projectives L_1, L_2 so that

$$M \oplus L_1 \cong N \oplus L_2.$$

As a corollary to 1.3 we can easily check that the only connected locally finitely generated projectives are actually free, so we can assume L_1, L_2 are free.

Corollary 1.5 (Anderson-Brown-Peterson [4]). M is stably equivalent to N if and only if there is an $\mathcal{A}_1$ -map $f: M \rightarrow N$ inducing isomorphisms of Sq^1 and Q homologies if M, N are connected and locally finite.

The stable equivalence classes of finitely generated $\mathcal{A}_1$ -modules form a ring $\mathcal{M}$ under direct sum and tensor product. The 0-module acts as additive unit and the module $\mathbb{Z}/2$ acts as the multiplicative unit. Adams and Priddy have extensively analyzed $\mathcal{M}$. However, we do not need their results here. What we need is the subring generated by $\overline{\mathcal{A}_1}$.

Here are minimal models for modules in the stable classes of $\underbrace{(\overline{\mathcal{A}_1}) \otimes \dots \otimes \overline{\mathcal{A}_1}}_{i\text{-times}}$ for some i .

Definition 1.6. Let Y_i be the $\mathcal{A}_1$ module with generators

$$y_0 y_4 y_8 \cdots y_{4i}$$

and relations $Sq^{2,1,2} y_{4(j-1)} = Sq^1 y_{4j}$, $Sq^1 y_0 = 0$, $Sq^{2,1,2} y_{4i} = 0$.

Here is a picture of Y_1

$$(1.7) \quad \begin{array}{c} \text{Sq}^2 \quad \text{Sq}^2 \quad \text{Sq}^2 \\ \text{Sq}^1 \quad \text{Sq}^1 \quad \text{Sq}^1 \end{array}$$

Again we have

Definition 1.8. Let z_i be the $\mathcal{A}_1$ module with generators $z_0, z_4, \dots, z_{4i}$ and relations

$$Sq^1(z_0) = 0, Sq^2(z_{4i}) = 0, Sq^{2,1,2} z_{4j} = Sq^1 z_{4(j+1)}$$

$$(1.9) \quad \begin{array}{c} \text{Sq}^2 \quad \text{Sq}^2 \quad \text{Sq}^2 \\ z_0 \quad z_1 \quad z_1 \quad z_2 \end{array}$$

Definition 1.10. (a) Let s_i be the $\mathcal{A}_1$ module with generators $s_0, s_4, \dots, s_{4i}$, h and with relations

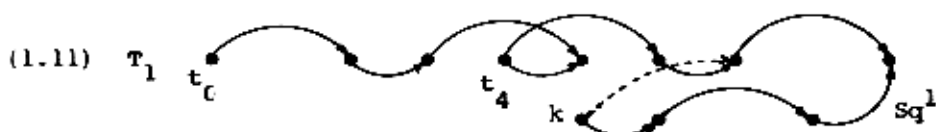
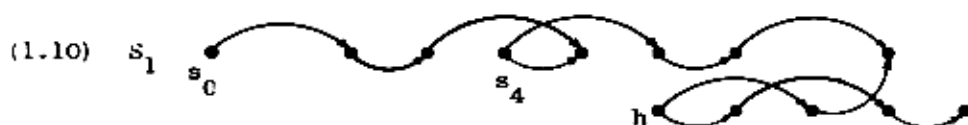
$$Sq^1(s_{4(r+1)}) = Sq^{2,1,2}(s_{4r}), Sq^1(s_0) = 0$$

$$Sq^3(h) = Sq^{2,1,2}(s_{4i}).$$

(b) Let T_i be the $\mathcal{A}_1$ -module with generators $t_0, \dots, t_{4i}, k$ with

$$Sq^1(t_{4(r+1)}) = Sq^{2,1,2}(t_{4r}), \quad Sq^1(t_0) = 0$$

$$Sq^3 t_{4i} = Sq^2 k.$$



Lemma 1.12. All four of these modules are minimal in their stable classes and any $\mathcal{A}_1$ -map $T_1 \rightarrow T_i$, $S_1 \rightarrow S_i$, $Z_1 \rightarrow Z_i$, $Y_1 \rightarrow Y_i$ which is non-trivial in degree 0 is an isomorphism.

Proof. From the relations $Sq^{2,1,2}(y_{4(r-1)}) = Sq^1 y_{4r}$ it follows that the y_i, z_i, s_i, t_i all map non-trivially. This is the proof for Y_i, Z_i . For k in T_i we see that k goes either to k or $k + Sq^1(t_i)$ and in either case the map is an isomorphism, similarly for S_i .

Lemma 1.13. (a) The Sq^1 homology of any of these modules is generated by the bottom class,

$$\begin{aligned} (b) \quad H_*(Y_i, \mathbb{Q}) &= \mathbb{Z}/2 \{Sq^2 y_{4i}\} \\ H_*(Z_i, \mathbb{Q}) &= \mathbb{Z}/2 \{z_{4i}\} \\ H_*(S_i, \mathbb{Q}) &= \mathbb{Z}/2 \{h + Sq^2(s_{4i})\} \\ H_*(T_i, \mathbb{Q}) &= \mathbb{Z}/2 \{Sq^1 k + Sq^2 t_{4i}\} \end{aligned}$$

(obvious).

Theorem 1.14. (a) Y_{2i+1} is a minimal model for $\bar{\mathcal{A}}^{4i+3}$.
 (b) Z_{2i} is a minimal model for $\bar{\mathcal{A}}^{4i}$.
 (c) T_{2i} is a minimal model for $\bar{\mathcal{A}}^{4i+1}$.
 (d) S_{2i} is a minimal model for $\bar{\mathcal{A}}^{4i+2}$.

(The proof is a tedious but routine construction. We take, for example, the obvious map of free $\mathcal{A}_1$ modules onto Z_{2i} and verify the kernel is T_{2i} . Similarly, for T_{2i} , S_{2i} , and Y_{2i+1} . Then check the first three stages of an actual resolution of $\mathbb{Z}/2$, note that the bar resolution has kernel $\bar{\mathcal{A}}^i$ at stage i and verify that the stable class of the kernel at stage i does not depend on the particular resolution.)

As an easy corollary we have

Corollary 1.15.

$$\begin{aligned} \{S_l\} &= \{Y_0 \odot Y_l\} \\ \{Y_l\} &= \{Y_0 \odot Z_l\} \\ \{T_{l+1}\} &= \{Y_0 \odot S_l\} \\ \{Z_{l+1}\} &= \{Y_0 \odot T_l\} \end{aligned}$$

and in general this set of stable modules is closed under tensor products.

2. The spaces associated to a resolution of $Z/2$ over $\mathcal{A}_1$

The map $\partial_0: \mathcal{B}_0 \rightarrow Z/2$. $\mathcal{B}_0$ can be written as $\mathcal{A}_1(e_1) + \dots + \mathcal{A}_1(e_n)$ and associated to ∂_0 we choose a map

$$d_0: b_0 \rightarrow \bigvee_1^n \Sigma^{\dim(e_r)} K(Z/2, 0)$$

defined by $d_0^*(e_r) = \partial_0(e_r)I$. The cofiber of d_0 and the associated map

$$M(d_0) \xrightarrow{k} \Sigma b_0$$

provides the first filtration space $b_0^{(1)}$ of b_0 . Next, associated to $\partial_1: \mathcal{A}_1(f_1) \oplus \dots \oplus \mathcal{A}_1(f_m) \rightarrow \Sigma \mathcal{A}_1(e_i)$ there is a map

$$d_1: \bigvee_1^n \Sigma^{\dim(e_r)} K(Z/2, 0) \rightarrow \bigvee \Sigma^{\dim f_j} K(Z/2, 0)$$

defined by

$$(2.1) \quad d_1^*(\iota_j) = \Sigma \langle \partial_1(f_j), e_k \rangle \iota_k.$$

Since $\partial_0 \partial_1 = 0$, $d_1 d_0 \approx 0$ and there is a lifting

$\ell: b_0^{(1)} \rightarrow \bigvee \Sigma^{\dim f_j} K(Z/2, 0)$. The cofiber of ℓ is $b_0^{(2)}$ and the composite

$$(2.2) \quad b_0^{(2)} \xrightarrow{k} \Sigma b_0^{(1)} \xrightarrow{\Sigma k} \Sigma^2 b_0$$

gives the second filtration of b_0 . We now iterate the procedure obtaining the sequence

$$(2.3) \quad \rightarrow \Sigma^{-i} b_0^{(i)} \rightarrow \Sigma^{-i+1} b_0^{(i-1)} \rightarrow \dots \rightarrow b_0$$

associated to our resolution of $\mathbb{Z}/2$ over $\mathcal{A}_1$.

Observe that the cohomology maps (mod 2) in (2.3) are all trivial and the cofiber at each stage is a wedge of suspensions of $K(\mathbb{Z}/2, 0)$. Hence (2.3) represents an Adams resolution of b_0 .

Theorem 2.4. Let the resolution of $\mathbb{Z}/2$ over $\mathcal{A}_1$ be minimal; then

$$H^*(b_0^{(4i+3)}) \cong \mathcal{A}(2)/\mathcal{A}(2) \mathcal{A}_1 \otimes Y_{2i+1}$$

$$H^*(b_0^{(4i)}) \cong \mathcal{A}(2)/\mathcal{A}(2) \mathcal{A}_1 \otimes Z_{2i}$$

as modules over $\mathcal{A}_1$

Indeed, it is easily seen by induction that, for example, as a module over $\mathcal{A}(2)$, $H^*(b_0^{(4i)})$ is given by

$$\mathcal{A}(2)(e_0) \oplus \mathcal{A}(2)(e_4) \oplus \dots \oplus \mathcal{A}(2)(e_{8i})$$

subject to relations $Sq^1(e_0) = 0$, $Sq^{2,1,2}(e_{4r}) = Sq^1(e_{4(r+1)})$,

and $Sq^2(e_{8i}) = 0$ and similarly for $H^*(b_0^{(4i-1)})$.

Corollary 2.5. For the minimal resolution

$$\text{Ext}_{\mathcal{A}(2)}^s H^*(b_0^j, \mathbb{Z}/2) \cong \text{Ext}_{\mathcal{A}_1}^{s+j}(\mathbb{Z}/2, \mathbb{Z}/2).$$

Thus, for example, in the case of $b_0^{(4)}$ the Ext groups have the form

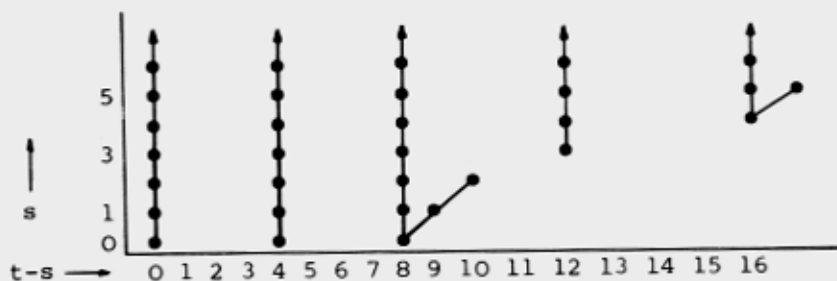


Figure 2.6

The Adams resolution (2.3) has certain universal properties. Thus, for example, by obstruction theory as in [8, Chapter 2] from the map we obtain the lifted maps

$$u_{ij} : b_0^{(i)} \wedge b_0^{(j)} \longrightarrow b_0^{(i+j)}$$

Moreover, for any other Adams resolution of b_0 we obtain similar maps and maps

$$\lambda_i \bar{b}_0^{(i)} \longrightarrow b_0^{(i)}$$

so the diagrams

$$\begin{array}{ccc}
 \bar{b}_0^{(i)} \wedge \bar{b}_0^{(j)} & \rightarrow & \bar{b}_0^{(i+j)} \\
 \downarrow & & \downarrow \\
 b_0^{(i)} \wedge b_0^{(j)} & \rightarrow & b_0^{(i+j)}
 \end{array}$$

commute up to filtration degree -1 homotopies.

3. The Adams map ψ^3

In the introduction we described how the Adams operation ψ^3 is defined in b_0 . Here we give some more explicit properties of certain polynomials in ψ^3 .

Definition 3.1. The Postnikov degree k of a map $\varphi: X \rightarrow X$ is the biggest Postnikov connective cover of X through which φ factors:

$$\begin{array}{ccc}
 & & X_k \\
 & \nearrow \tilde{\varphi} & \downarrow (j) \\
 X & \xrightarrow{\varphi} & X
 \end{array}$$

Definition 3.2. The Adams degree s of φ is the largest Adams filtration of X through which φ factors:

$$\begin{array}{ccc}
 & & X^{(s)} \\
 & \nearrow \tilde{\varphi} & \downarrow \\
 X & \xrightarrow{\varphi} & X
 \end{array}$$

Example 3.3. $\psi^3 - 1$ has Adams and Postnikov degrees 3 and 4, respectively, hence factors through a map θ :

$$\begin{array}{ccc} & & \Sigma^4 b_{sp} \\ & \nearrow \theta & \downarrow i \\ b_0 & \xrightarrow{\psi^3 - 1} & b_0 \end{array}$$

Lemma 3.4. $\theta^*(I_{sp}) = Sq^4(I)$.

Proof. $(\psi^3 - 1)(q) = 8q$. Hence in dimension 4, $(\psi^3 - 1)^*$ is multiplication by 8. On the other hand, the cohomology map of a torsion free class to the i^{th} stage of any Adams resolution takes the generator to 2^i times the corresponding generator in X^i . Thus i^* is already multiplication by 8 in dimension 4 so (3.4) follows.

Next consider the operation φ_n defined in the introduction.

Lemma 3.5. φ_n^* is 0 in homotopy in dimensions $< 4n$ and in dimensions $\geq 4n$ is multiplication by λ_i with each λ_i divisible by $2^{4n-\alpha(n)}$ and $\lambda_{4n} = 2^{4n-\alpha(n)}$ (odd).

In particular, this shows that the Adams filtration of $\varphi_n \leq 4n - \alpha(n)$ and the Postnikov filtration $\leq 4n$.

Proof. The generator in dimension $4j$ in homotopy g_j is mapped by ψ^3 to $9^j g_j$, hence

$$\begin{aligned}\varphi_n(g_j) &= (9^j-1) \dots (9^j-9^{n-1})g_j \\ &= (9^j-1)(9^{j-1}-1) \dots (9^{j-n+1}-1)9^r(g_j)\end{aligned}$$

which equals 0 if $j \leq n-1$. Also on the torsion ψ^3 is the identity so $\varphi_n(\text{torsion}) = 0$. Now, it is well known [2] that $(9^i-1) = 2^{3+v_2(i)}$ (odd) where $v_2(i)$ is the exponent of 2 in the prime factorization of i. Thus the exponent of 2 in dimension $4i$ is $(v_2(i)+3) + (v_2(i-1)+3) + \dots + (v_2(i-n+1)+3)$ but $v_2(i) + \dots + v_2(i-n+1) = v_2(i!/(i-n)!)$. Now $r! = 2^{r-\alpha(r)}$ (odd) so

$$v_2(i!/(i-n)!) = n-\alpha(i)+\alpha(i-n)$$

which takes its minimum when $i = n$ since $\alpha(n)+\alpha(m) \geq \alpha(n+m)$ for all positive n, m .

Now Theorem B amounts to proving that these bounds are best possible. Special circumstances gave the result for ψ^3-1 since the groups in which obstructions could appear were all zero, but to prove the general result we cannot hope to argue one stage at a time. Instead it will depend on making extensive calculations with Adams spectral sequences.

4. Mahowald's theorem and the homotopy type of $b_0 \wedge b_0$

In this section we recall Mahowald's theorem [7] giving the

structure of $H^*(b_0, \mathbb{Z}/2)$ as a module over $\mathcal{A}_1$ and using it and Theorem B gives the proof of Theorems A and C.

Mahowald's result depends on

Theorem 4.1 (Anderson, Brown, Peterson [4]).

$$H^*(\mathcal{A}(2)/\mathcal{A}(2) \overline{\mathcal{A}}_1, Sq^1) \cong P(\xi_1^4)$$

$$H^*(\mathcal{A}(2)/\mathcal{A}(2) \overline{\mathcal{A}}_1, \mathbb{Q}) \cong E(\xi_2^2, \xi_3^2, \dots, \xi_n^2, \dots).$$

Proof. Let $\mathcal{A}(2)^*$ be given as $P(\xi_1, \dots, \xi_n, \dots)$ with

$\psi(\xi_i) = \sum \xi_j \otimes \xi_{n-j}^{2^{n-j}}$. Then the $\mathbb{Q}$ homology can be calculated by

dualizing $\langle \bar{Q}, \xi^I \rangle = \sum \langle \bar{Q}, \xi_1^I \rangle \otimes \xi_2^I$ where $\psi \xi^I = \sum \xi_1^I \otimes \xi_2^I$. This is

a derivation and on generators $\bar{Q}(\xi_i) = \xi_{i-2}^2, i > 2, \bar{Q}(\xi_2) = 1$.

But as a coalgebra over $\mathcal{A}(2)^*$, $H_*(b_0, \mathbb{Z}/2) = P(\xi_1^4, \xi_2^2, \xi_3^2, \dots)$.

The second statement now follows. Similarly for the first

$Sq^1(\xi_i) = \xi_{i-1}^2, i > 1, Sq^1(\xi_1) = 1$ and this again gives a derivation so (3.1) follows.

Next we have

Theorem 4.2 (Mahowald). There exist $\mathcal{A}_1$ modules M_{2^i} ,

$i \geq 2$, and surjections $\phi_i : \mathcal{A}(2)/\mathcal{A}(2) \overline{\mathcal{A}}_1 \rightarrow \Sigma^{2^i} M_{2^i}$ satisfying

$$(1) \quad M_{2^i} \cong Y_{2^{i-2}-1}$$

$$(2) \quad H_*(\mathcal{A}(2)/\mathcal{A}(2) \overline{\mathcal{A}}_1, Sq^1) \xrightarrow{\phi_*} H_*(M_{2^i}, Sq^1).$$

$$H_*(\mathcal{A}(2)/\mathcal{A}(2)\overline{\mathcal{A}}_1, \mathbb{Q}) \xrightarrow{\varphi_*} H_*(M_{2^i}, \mathbb{Q})$$

are surjections with $\varphi^*(g_{Sq^1}) = \xi_1^{2^i}$, $\varphi^*(g_Q) = \xi_i^{2^i}$.

Proof. Again we work with the dual algebra. Consider the sequence

$$\begin{aligned} &\xi_{i+1}^2, \xi_i^2, \xi_i \xi_{i-1}^2, \xi_{i-1}^4, \xi_i \xi_{i-2}^4, \xi_{i-1}^2 \xi_{i-2}^4, \dots, \xi_{i-r} \xi_{i-r-2}^{4j} \xi_{i-r-1}^{2s}, \\ &\xi_{i-r-2}^{4j} \xi_{i-r-1}^{2s+2}, \xi_{i-r} \xi_{i-r-2}^{4(j+1)} \xi_{i-r-1}^{2s-2}, \xi_{i-r-2}^{4(j+1)} \xi_{i-r-1}^{2s}, \dots, \xi_{i-r-2}^{2^{r+3}} \dots \end{aligned}$$

terminating in

$$\xi_3 \xi_2^2 \xi_1^{2^{i-8}}, \xi_2^4 \xi_1^{2^{i-8}}, \xi_3 \xi_1^{2^{i-4}}, \xi_2^2 \xi_1^{2^{i-4}}, 0, \xi_1^{2^i}.$$

The sequence is closed under $\overline{Sq}^1$. We show it also closed under $\overline{Sq}^2$. Note $\overline{Sq}^2(ab) = (\overline{Sq}^2 a)b + a(\overline{Sq}^1 b) + (\overline{Sq}^1 a)(\overline{Sq}^1 b)$; hence $\overline{Sq}^2(\xi_i) = 0$, $\overline{Sq}^2(\xi_i^{2s}) = s(\xi_{i-1}^4 \xi_i^{2s-2})$, and from these two formulas the result follows.

Now (4.2) may be extended. Consider, for example, the composite $\varphi_{ij}: \mathcal{A}/\mathcal{A}\overline{\mathcal{A}}_1 \xrightarrow{\psi^*} \mathcal{A}/\mathcal{A}\overline{\mathcal{A}}_1 \otimes \mathcal{A}/\mathcal{A}\overline{\mathcal{A}}_1 \rightarrow M_{2^i} \otimes M_{2^j}$. Clearly, $\varphi_{ij}^*(g \otimes g_{Sq^1}) = \{\xi_1^{2^i} \cdot \xi_1^{2^j}\}$, $\varphi_{ij}^*(g \otimes g_Q) = \{\xi_i^2 \xi_j^2\}$. More generally, on iterating we obtain $\mathcal{A}_1$ -maps

$$\varphi_{i_1, i_2, \dots, i_r}: \mathcal{A}/\mathcal{A}\overline{\mathcal{A}}_1 \rightarrow M_{2^{i_1}} \otimes \dots \otimes M_{2^{i_r}} \text{ with}$$

$$\phi_{i_1 \dots i_r}^* (g \otimes \dots \otimes g_{Sq^1}) = \left\{ \xi_{i_1}^{2^{i_1}} \dots \xi_{i_r}^{2^{i_r}} \right\}$$

$$\phi_{i_1 \dots i_r}^* (g \otimes \dots \otimes g_Q) = \left\{ \xi_{i_1}^{2^{i_1}} \dots \xi_{i_r}^{2^{i_r}} \right\}.$$

Of course, we may substitute for these tensor products their minimal stable $\mathcal{A}_1$ -module representatives, and we assume that done. Consequently we have

Theorem 4.3 (Mahowald). There is a free $\mathcal{A}_1$ -module F and an isomorphism of $\mathcal{A}_1$ -modules

$$F \oplus \sum_{1 \leq i_1 < \dots < i_r} \Sigma^{2^{i_1 + \dots + i_r}} \left\{ Y_{2^{i_1-2}-1} \otimes \dots \otimes Y_{2^{i_r-2}-1} \right\} \\ \approx \mathcal{A}(2) / \mathcal{A}(2) \overline{\mathcal{A}}_1.$$

Using (4.3) together with Theorem B we prove Theorems A and C as follows. Note that in the minimal resolution of $\mathbb{Z}/2$ the corresponding space $b_0^{(n)}$ have $\mathcal{A}_1$ -module structure $\{\mathcal{A}^n\} \otimes \mathcal{A} / \mathcal{A} \overline{\mathcal{A}}_1$ where $\mathcal{A}^n$ is represented by its minimal model. Moreover, an easy calculation shows that

$$\left\{ \begin{array}{l} \theta_1 : b_0 \rightarrow \Sigma^4 b_{sp} \\ \theta_{2^{i-2}} : b_0 \rightarrow \Sigma^{2^i} b_0^{(2^{i-2}-1)} \end{array} \right\} \text{ satisfies } \theta_{2^{i-2}}^* (\bar{g}_{Sq^1}) = (\bar{\xi}_1^{2^i}),$$

Dually detect θ_{2^i}

since this latter class is the restriction of the integral generator and the calculation in (3.5) shows $\theta^*(\iota_{2i}) = (\text{odd})$ (integral generator). But from this an easy calculation shows

$\theta_{2i-2}(g_Q) = (\xi_i^2)$. Next we construct maps

$$\theta_{2n} : b_0 \longrightarrow \Sigma^{2n} b_0^{4n-\alpha(n)}$$

and similarly the θ_{2n+1} . For example, to construct θ_6 use the composite

$$(4.4) \quad b_0 \xrightarrow{\theta_4} \Sigma^{16} (b_0^7) \xrightarrow{\hat{\theta}_2} \Sigma^{24} b_0^{(10)}$$

where $\hat{\theta}_2$ is the lifting in the Adams filtration of θ_2 through seven stages. Next we construct out of these maps the desired map $b_0 \wedge b_0 \longrightarrow \bigvee \Sigma^{8n} b_0^{4n-\alpha(n)} \bigvee \Sigma^{8n+4} b_{sp}^{4n-\alpha(n)}$ using the composites

$$b_0 \wedge b_0 \xrightarrow{\theta_{2n} \wedge 1} \Sigma^{8n} b_0^{4n-\alpha(n)} \wedge b_0 \longrightarrow \Sigma^{8n} b_0^{4n-\alpha(n)}$$

Then the calculations above show μ^* for Sq^1 and Q homology are both isomorphisms. Also by minimality μ^* is an injection in cohomology with $\mathcal{A}(2)$ -free cokernel. We extend now to a homotopy equivalence by simply mapping into $\Sigma^n(K(Z/2, 0))$'s 1 for each free generator in the cokernel.

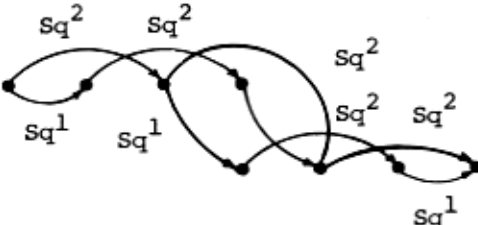
5. The proof of Theorem B

We begin by calculating $\text{Ext}_{\mathcal{A}(2)}^{S^*}(Z/2, H^*(Db_0 \wedge b_0))$ for

$s > 0$. Here Db_0 is the Spanier-Whitehead dual of b_0 .

Lemma 5.1. If as a module over $\mathcal{A}_1$, $H^*(X) = M_1 \oplus M_2$, then as a module over $\mathcal{A}_1$, $H^*(D(X)) = D(M_1) \oplus D(M_2)$. Moreover, the dual of $\mathcal{A}_1$ is again $\mathcal{A}_1$.

Proof. As the splitting is clear we need only show the second statement. We write $\mathcal{A}_1$

(5.2) 

This picture is symmetric and that is the result.

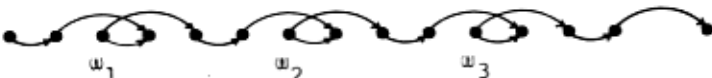
Lemma 5.3. (a) The dual of Y_i has generators $\theta_0, w_1, \dots, w_i$ with relations

$$Sq^{2,1}\theta_1 = Sq^1\theta_2, \dots, Sq^{2,1,2}w_{j-1} = Sq^1w_j, \quad j < i.$$

(b) The dual of Z_i has generators $z_0, \dots, z_{i-1}$ with relations

$$Sq^{2,1}z_1 = 0, \quad Sq^{2,1,2}(z_{j-1}) = Sq^1z_j, \quad j < i-1.$$

Proof. (a) The module described has picture

(5.4) 

which is clearly the reverse of Y_i . (b) Is similar.

Lemma 5.5. Let L_i be the module with generators
 $e_0, e_1, \dots, e_i$ and relations

$$Sq^{2,1,2}(e_j) = Sq^1(e_{j+1}), \quad j < i;$$

then there are exact sequences

$$(a) \quad 0 \rightarrow \Sigma M \rightarrow D(T_i) \rightarrow L_i \rightarrow 0,$$

$$(b) \quad 0 \rightarrow N \rightarrow D(S_i) \rightarrow \Sigma L_i \rightarrow 0,$$

where

$$M = \mathcal{A}_1 / \mathcal{A}_1 (Sq^1, Sq^{2,1,2}) = Y_0$$

$$N = \mathcal{A}_1 / \mathcal{A}_1 Sq^3.$$

Proof. Turn diagrams (1.10), (1.11) on end; then N and M are the modules on the second lines in each case.

Remark 5.6. The extensions are given by $Sq^1(e_0) = \Sigma I$ in (a) and $Sq^1(\Sigma e_0) = Sq^2 I$ in (b).

Similarly we have

Lemma 5.7. (a) There is an exact sequence

$$0 \rightarrow P \rightarrow D(Y_i) \rightarrow \Sigma^2 L_{i-1} \rightarrow 0$$

where

$$P = \mathcal{A}_1 / \mathcal{A}_1 Sq^2.$$

(b) There is an exact sequence

$$0 \rightarrow \Sigma(Z/2) \rightarrow D(Z_i) \rightarrow L_{i-1} \rightarrow 0$$

and the extensions are given by $Sq^1 e_0 = Sq^{2,1} \Sigma I$ in (a),

$Sq^1(e_0) = \Sigma I$ in (b). (From (5.3).)

Lemma 5.8. $\text{Ext}_{\mathcal{A}_1}^{**}(L_i, Z/2)$ has generators $e_0, \dots, e_i$ and relations

$$qe_r = h_0^3 e_{r+1}$$

$$h_1 e_r = 0$$

all r . In particular, $\text{Ext}_{\mathcal{A}_1}^{s,t}(L_i, Z/2) = 0$ for $t-s > 4i$.

Proof. There is an exact sequence

$$\mathcal{A}_1 / \mathcal{A}_1 Sq^1 \rightarrow L_i \rightarrow \Sigma^4 L_{i-1},$$

consequently a long exact sequence of Ext groups. Note using change of rings $\text{Ext}_{\mathcal{A}_1}(\mathcal{A}_1 / \mathcal{A}_1 Sq^1, Z/2) = \text{Ext}_{E(Sq^1)}(Z/2, Z/2) =$

$P(h_0)e_0$. Induction produces all of (5.8) except $qe_r = h_0^3 e_{r+1}$.

But through a range $L_i \cong \{\bar{A}^{4i}\}$ and the desired relation holds for $\text{Ext}_{\mathcal{A}_1}^{s,t}(\bar{A}^{4i}, Z/2) \cong \text{Ext}_{\mathcal{A}_1}^{s+4i,t}(Z/2, Z/2)$, $s > 0$.

Now from (5.1) we have

Theorem 5.9. As a module over $\mathcal{A}_1$

$$\begin{aligned}
 H^*(D(b_0), \mathbb{Z}/2) &\cong S^0 \bigvee_{\alpha(j) \equiv 0(4)} \Sigma^{(-8j+2\alpha(j)-1)} D(\mathbb{Z}_{(j-\alpha(j)/2)}) \\
 &\quad \bigvee_{\alpha(j) \equiv 1(4)} \Sigma^{-8j+2\alpha(j)-1} D(\mathbb{Y}_{(j-(\alpha(j)+1/2))}) \\
 &\quad \bigvee_{\alpha(j) \equiv 2(4)} \Sigma^{-8j+2\alpha(j)-2} D(S_{(j-\alpha(j)/2)}^{-1}) \\
 &\quad \bigvee_{\alpha(j) \equiv 3(4)} \Sigma^{-8j+2\alpha(j)-3} D(T_{(j-(\alpha(j)+1/2))}) \\
 &\quad \bigvee_F.
 \end{aligned}$$

Here $j = 1, 2, 3, \dots$, and F is a free $\mathcal{A}_1$ -module. Also, $D(\)$ has its lowest generator in dimension 0.

Thus, ignoring the free part the first few terms are

$$(5.10) \quad (S^0) \vee \Sigma^{-7} D(Y_0) \vee \Sigma^{-15} D(Y_1) \vee \Sigma^{-22} D(S_1) \vee \Sigma^{-31} D(Y_3) \vee \dots$$

Proof. Let $j = 2^{i_1} + \dots + 2^{i_r}$, then stably

$$(5.11) \quad Y_{(2^{i_1-1})} \otimes \dots \otimes Y_{(2^{i_r-1})} \cong \begin{cases} Z_{j-\frac{1}{2}r} & r \equiv 0(4) \\ Y_{j-\frac{1}{2}(r+1)} & r \equiv 1(4) \\ S_{j-\frac{1}{2}(r+2)} & r \equiv 2(4) \\ T_{j-\frac{1}{2}(r+1)} & r \equiv 3(4) \end{cases}$$

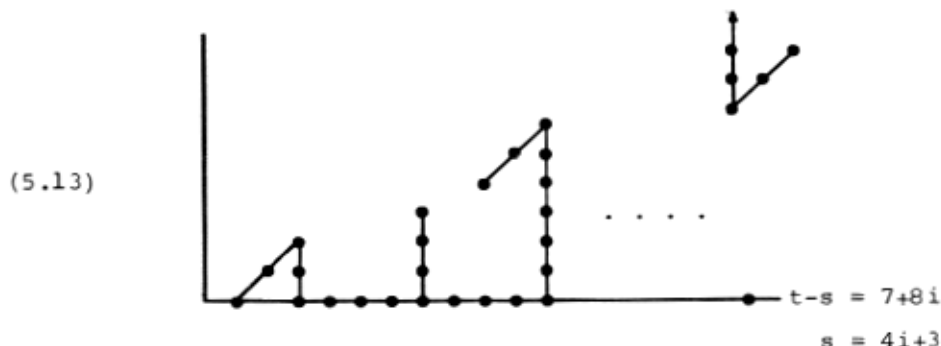
Now, the dimension of Z_k is $4k+1$, that of Y_k is $4k+3$, that of S_k is $4k+6$, that of T_k is $4k+5$. Thus, for example, the dual of

$$\begin{aligned} \Sigma^{4j} Z_{j-\frac{1}{2}\alpha(j)} &\cong \Sigma^{-4j-4(j-\frac{1}{2}\alpha(j))-1} D(Z_{(j-\frac{1}{2}\alpha(j))}) = \\ &= \Sigma^{-8j+2\alpha(j)-1} D(Z_{j-\frac{1}{2}\alpha(j)}) \end{aligned}$$

and these are the first wedge summands in (4.11). The others are handled similarly.

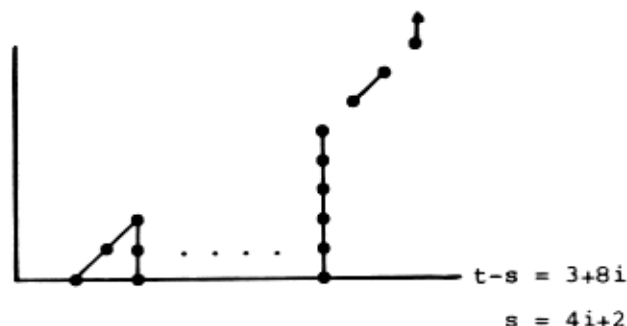
We next need

Lemma 5.12. (a) $\text{Ext}_{\mathcal{A}_1}(D(Y_{2i+1}, \mathbb{Z}/2))$ has the form



(b) $\text{Ext}_{\mathcal{A}_1}(D(Y_{2i}), \mathbb{Z}/2)$ has the form

(5.14)



Proof. Immediate from (5.7), (5.8).

Similar calculations can be made in the same way for $D(\mathbb{Z})$, $D(S)$, $D(T)$. We leave the details to the reader.

What we need to observe is that after suspending the correct number of times to put this into $DH^*(b_0)$ the first spike generator for the piece $\{D\Sigma^{8r}(X_r)\}$ occurs in $t-s = -8r$ and in s filtration $4r - \alpha(r)$. Similarly, the first spike for $D(\Sigma^{8r+4}(W_r))$ occurs in $t-s = -8r-4$ and s filtration $4r - \alpha(r)$, and since $f(r) = (4r - \alpha(r))$ is a monotone increasing function of r it is easily checked that these generating spikes and indeed all the spikes represent infinite cycles in the Adams spectral sequence.

To complete the proof we need

Theorem 5.15. Let L, M be connected modules of locally-
finite type over $\mathcal{A}(2)$; then

$$\text{Ext}_{\mathcal{A}(2)}(L, M) \cong \text{Ext}_{\mathcal{A}(2)}(D(M) \otimes L, \mathbb{Z}/2).$$

Proof. Let

$$\cdots \rightarrow C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow L \rightarrow 0$$

be an (2) free resolution of L which we may also suppose to be of locally finite type. Then $\text{Ext}(L, M)$ is the homology of the complex

$$0 \rightarrow \text{Hom}_{\mathcal{A}}(C_0, M) \xrightarrow{\delta_1} \text{Hom}_{\mathcal{A}}(C_1, M) \rightarrow \dots$$

On the other hand, we have

Lemma 5.16. $\text{Hom}_{\mathcal{A}}(L, M) \cong \text{Hom}_{\mathcal{A}}(D(M) \otimes L, \mathbb{Z}/2)$ where the isomorphism is explicitly given by the formula

$$h(f)(g \otimes l) = \langle g, f(l) \rangle.$$

(Here we regard $D(M)$ as $\text{Hom}(M, \mathbb{Z}/2)$ with $\mathcal{A}$ -module action given by $\alpha(g)(m) = g(\chi(g)m)$ where χ is the canonical anti-automorphism.)

Proof. $h(f) \in \text{Hom}_{\mathbb{Z}/2}(D(M) \otimes L, \mathbb{Z}/2)$ is certainly well defined. We must check that it is an $\mathcal{A}(2)$ -map. Notice

$$\begin{aligned} h(f) \alpha(g \otimes l) &= h(f) \Sigma \alpha_i(g) \otimes \alpha'_i(l) \\ &= \Sigma \langle \alpha_i(g), f \alpha'_i(l) \rangle. \end{aligned}$$

By assumption f is an $\mathcal{A}$ -map so this last sum is

$$\begin{aligned}\Sigma(\alpha_i(g), \alpha_i' f(l)) &= \Sigma(g, \chi(\alpha_i) \alpha_i' f(l)) \\ &= \langle g, (\Sigma \chi(\alpha_i) \alpha_i') f(l) \rangle\end{aligned}$$

but $\Sigma \chi(\alpha_i) \alpha_i' = 0$ for $\alpha \neq 1$ so this is $\epsilon(\alpha) \langle g, f(l) \rangle$ where ϵ is the augmentation. But this is by definition the action of $\mathcal{A}$ on $Z/2$.

We now define a map

$$(5.17) \quad k : \text{Hom}_{\mathcal{A}}(M \otimes L, Z/2) \longrightarrow \text{Hom}_{\mathcal{A}}(M, \text{Hom}_{Z/2}(L, Z/2))$$

by the rule

$$\langle (kr)(m), l \rangle = \langle r, m \otimes l \rangle.$$

We verify that k is indeed an $\mathcal{A}$ -map

$$\begin{aligned}\langle [k(r)\alpha + \alpha k(r)](m), l \rangle &= \\ &= \langle r, (\alpha m) \otimes l \rangle + \langle r, (m \otimes (\chi(\alpha)l)) \rangle = \\ &= \langle \alpha m \otimes l + m \otimes \chi(\alpha)l \rangle\end{aligned}$$

and in general

$$\alpha(m) \otimes l + m \otimes \chi(\alpha)l$$

is decomposable over $\overline{\mathcal{A}} = \ker(\epsilon)$ [5]. Hence since $r(\theta(x)) = \epsilon(\theta)r(x) = 0$ for $\theta \in \overline{\mathcal{A}}$ it follows that $k(r)\alpha + \alpha k(r) = 0$ all

so k is indeed an $\mathcal{A}$ -map.

The proof of (5.16) is now completed by (routinely) verifying that $h \cdot k = 1$ restricted to $D(M) \otimes L$ and checking that $kTh = 1$ on embedding

$$M \subset \text{Hom}_{Z/2}(\text{Hom}_{Z/2}(M, Z/2), Z/2)$$

The proof of (5.15) is now completed on noting that given an $\mathcal{A}$ -map $f: L \rightarrow L'$ there is the natural map

$$(5.18) \quad \text{Hom}_{\mathcal{A}}(L', M) \xrightarrow{D(f)} \text{Hom}_{\mathcal{A}}(L, M)$$

and the diagram

$$\begin{array}{ccc} \text{Hom}_{\mathcal{A}}(L', M) & \xrightarrow{D(f)} & \text{Hom}_{\mathcal{A}}(L, M) \\ \downarrow h \approx & & \downarrow h \\ \text{Hom}_{\mathcal{A}}(D(M) \otimes L', Z/2) & \xrightarrow{D(1 \otimes f)} & \text{Hom}_{\mathcal{A}}(D(M) \otimes L, Z/2) \end{array}$$

commutes. On applying this to the chain complex defining $\text{Ext}_{\mathcal{A}}(L, M)$ we see at once that it is isomorphic to the chain complex defining $\text{Ext}_{\mathcal{A}}(D(M) \otimes L, Z/2)$ and the result follows.

The results on $\text{Ext}_{\mathcal{A}(2)}^*(H^*(b_0), H^*(b_0))$ obtained above (5.9-5.14) give in particular

5.20. $\text{Ext}^{**} = 0$ if $t-s = -1$, and for $t-s = 0$, $\text{Ext}^{s,s}$ is a free module over $P(h_0)$ having generators $e_0, e_1, e_2, \dots$

where the s-degree of e_i is $4i - \alpha(i)$.

We now complete the proof of Theorem B. It has already been observed that the spikes are infinite cycles. Now, the φ_n are all linearly independent (since $\varphi_{i*} = 0$ in dimensions less than $4i$ but not in $4i$). φ_1 has s-degree 3 and $\varphi_2 = a + b\varphi_1 + \tau$ where by (5.20) τ must have s-degree ≥ 7 . Checking on homotopy in dimension zero, $\tau_*(I) + aI = 0$. But $\tau_*(I)$ must be divisible by 2^7 , hence $a = 2^7 a'$. Again in dimension 4

$$0 = 2^7 a' + 8b + \tau_*(q).$$

Here since q has filtration 3, $\tau_*(q) = 2^4 b'(q)$ and we see that b is divisible by 2^4 . Hence φ_2 actually has filtration 7 and we may repeat the argument with φ_3 , etc. This shows the φ_i lift to the correct Adams filtrations. We must still show their Postnikov filtrations are correct.

For example, we check on φ_2 . Note that we can choose $e_0, e_1, \dots$ so that $e_i = p(f_i)$ for $i > 1$ in Ext. Now multiplication by p corresponds to composition with the periodicity map in homotopy. Hence, having seen that e_0, e_1 are represented by 1, φ_1 we note that τ in the argument above can be chosen so $\tau = p(f)$. With this modification $\tau_*(I) = \tau_*(q) = 0$ so a and b in the argument above also $= 0$ and $\varphi_2 = p(f)$. Next we show φ_3 represented by this composite

$$b_0 \xrightarrow{\varphi_1} b_{sp} \xrightarrow{\hat{\varphi}_2} \mathfrak{g}_{sp}^{8b(3)}$$

and the remaining generators $e_4, e_5, \dots$ can be assumed to lie in the image of p^2 , etc.

Theorem B follows.

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